

Bright Learn Data Analytics

Research Assignment 1

Question 1 What is a Database? Define it in your own words and explain why databases are essential in the modern world. Provide two real-world examples of where databases are used.

Definition: A database serves as a digital hub designed to store large amounts of information securely, keeping everything highly organized and instantly searchable.

This is why we need databases today:

- Keep confidential information safe: Tight security and encryption processes are used to protect private information, which helps companies comply with laws such as GDPR or the Protection of Personal Information Act.
- Minimizes mistakes or errors: Prevents people from making duplicates, losing files and messing up data.
- Easy to find data: Information can be pulled up using one specific detail from billions of data in a split second.
- Ready for expansion: Databases handle large amounts of business data without slowing down or breaking.
- Seamless teamwork: Multiple colleagues can modify the same information at the same time without locking others out or crashing the system.

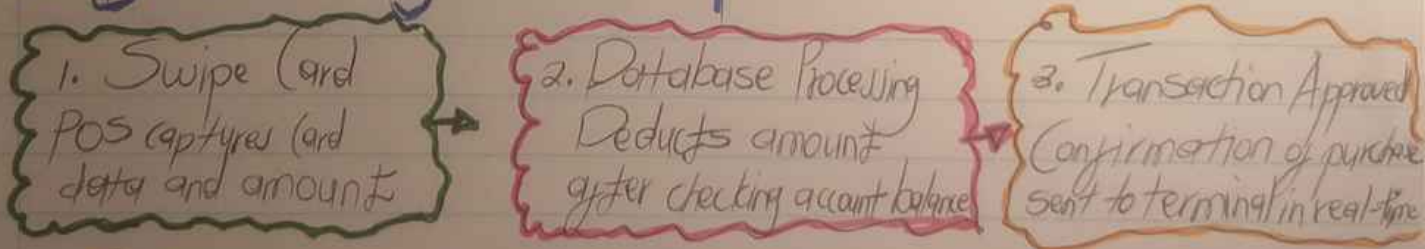
To summarize, databases are a lot more than just digital filing cabinets. They are basically what keeps the modern world running smoothly in the background. Whether you are swiping your card or checking your banking app, there's a database doing all the heavy lifting to keep that information secure, organized and moving fast. Without databases, the quick and reliable digital platforms we take for granted every day would just completely break down.

Here are two real-world examples of where databases are used.

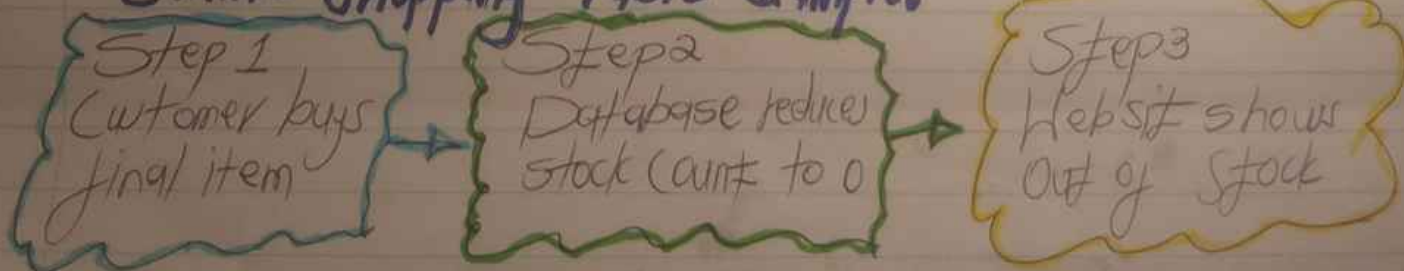
a) In a banking industry, specifically digital banking. Think about what happens when you tap or swipe your card at a till or log into your banking app. Your request basically shoots down a straight line to see if you have enough money and log the transaction. Behind the scenes, a database instantly checks your funds, stamps the exact second you made the purchase, and updates your balance everywhere so that you don't end up accidentally overdrawing your account.

b) Online shopping platform such as Takealot is also a perfect example. The moment you click "buy" online, a quick chain reaction kicks off to drop the stock count and lock in your order. Databases are constantly managing live inventory, prices and shipping addresses in the background. If you happen to snap up the very last item in stock, the database instantly refreshes the page for everyone else to see that it's sold out, making sure that nobody else can add it to their cart.

Digital Banking table example:



Online Shopping table example:



Question 2): What is a Data Warehouse? How does it differ from a regular database? Why would a company use a data warehouse?

Definition: According to Holdsworth, Kosinski and Jonker (2006), a data warehouse essentially "aggregates data from various sources into a central data store optimized for querying and analysis."

How a regular database differs from a data warehouse:

- What they actually do: A standard database is basically a fast cash register. Its only job is handling instantly, day-to-day things like recording the exact moment a customer buys a shirt. A data warehouse is completely different; it's more like a quiet corporate archive built to look at the bigger picture, like tracking how sales have shifted over 5 years.

- The way data behaves: The information in a regular database changes from one second to the next because it only cares about what's happening right now.

Warehouses hold data that stays frozen. It serves as a permanent historical record so that company analysts can look back and find accurate patterns without the data shifting around under them.

- Speed and design: Normal databases are built to save and process single entries one by one, lightning fast. Warehouses are built from the ground up for a totally different task, ~~combining~~ combing through and querying millions of rows of data all at once without lagging out the system.

Feature	Standard Database	Data Warehouse
Main Job	Logs quick daily actions, like selling a shirt.	Stores long-term trends over months or years.
The Data	Changes constantly, shows only right now.	Permanent record, never gets overwritten.
Speed	Saves single entries instantly.	Searches millions of rows all at once.
Analogy	A fast-moving cash register.	A quiet corporate archive.

Why Companies Use One?:

For starters, a warehouse makes long-term planning a whole lot easier for management. Instead of guessing, they can look back at years of historical data to figure out future trends and see how customer habits are shifting.

It also helps get every department onto the same page. Normally, you have messy, conflicting information floating around in sales, marketing and HR, which leads to teams constantly arguing over whose numbers are actually right. Putting everything in a single, clean setup gives everyone one reliable source of truth.

Lastly, it stops your everyday business apps from lagging. Running heavy reports on a live database can easily slow things down or crash the system, so moving that workload to a warehouse keeps everything running smoothly.

Question 3): What is a Data Lake? Explain the concept and describe how it differs from data warehouse. When would a business prefer a data lake over a data warehouse?

Definition: A data lake can be defined as a "centralized repository that allows you to store all your structured and unstructured data at any scale" (Amazon Web Services, 2026).

The concept of Data Lake: A data lake can be conceptualized as a massive, centralized digital storage tank. As detailed by MongoDB (2026), this architecture pulls in vast quantities of both current and historical data from completely different sources, safely holding onto it in its raw, original state, and form. The big selling point here is architectural flexibility. MongoDB (2026) highlights that unlike rigid traditional systems, you can dump structured, semi-structured, or totally unstructured files such as JSON scripts, CSV logs, PDFs and even audio files straight into a lake without doing any upfront formatting or schema planning. Furthermore, the platform provider notes that while companies usually set these environments up to feed advanced machine learning models and heavy analytics, they are also incredibly popular simply as a cheap, scalable place to park data for business projects that haven't even been thought of yet [MongoDB, 2026].

Key Differences between Data Lake and Data Warehouse:

When mapping out modern data infrastructure, the distinction between a data lake and a data warehouse fundamentally hinges on data maturity and structural intent. David (2028) asserts that a data warehouse functions

as a highly rigid, centralized repository optimized strictly for processed, clean, and highly structured data used in business intelligence applications. Conversely, the author details that a data lake serves as a massive, flexible storage system explicitly designed to hoard vast volumes of raw, completely unprocessed assets spanning everything from structured rows to unstructured social media feeds and streaming IoT logs [David, 2026].

The core operational friction points between these 2 environments typically surface across several distinct technical categories:

- The Blueprint Strategy: A data warehouse implements a strict "schema-on-write" philosophy, meaning engineers must scrub, transform, and map data before it can be loaded. In stark contrast, David (2026) observes that data lakes rely on a "schema-on-read" framework, which leaves the information completely untouched in its native state until someone actually runs an analytical query.

- Performance VS. Scalability: Because warehouses are heavily structured and tightly indexed, they provide incredibly fast SQL query times for executive dashboards but scale vertically at a much higher cost.

The analysis clarifies that data lakes compromise on raw query speeds due to their lack of structured formatting, but they offer immensely superior horizontal scalability by leveraging low-cost cloud object storage [David, 2026].

- Target Audience and AI Integration: Traditional warehouses are built primarily for business analysts running predefined KPI reports and standard corporate spreadsheets. Meanwhile, David (2026) highlights that data lakes act as the preferred playground for data scientists and engineers because their raw

formats are ideally suited for heavy data experimentation, AI modeling, and machine learning workloads.

Tactical Comparison Matrix:

As outlined by David (2026), the structural and financial trade-offs separating these platforms can be broken down across the following dimensions:

Comparison Aspect	Data Warehouse	Data Lake
a) Data Types Allowed	Structured tables, rows and columns only [David, 2026].	Structured, semi-structured, and unstructured files alike.
b) Processing Style	Strict ETL pipeline execution.	ETL or direct raw dumping [David, 2026].
c) Financial Overhead	Higher infrastructure cost due to tight compute coupling.	Drastically lower costs via cheap cloud object repositories.
d) Information Freshness	Periodic batch processing (typically hourly or daily updates).	Real-time or near-real-time streaming updates supported [David, 2026].

When would businesses or a business prefer a data lake over a data warehouse?

Businesses dealing with massive volumes of raw, unstructured data often prefer data lakes over warehouses to support rapid ingestion and advanced data science exploration [Workday, 2026]. When deciding between the 2 architectures, the choice largely depends on specific organizational requirements across several key operational dimensions. For main use cases, a data lake is ideal for building AI models, data science exploration, and parsing IoT feeds. In contrast,

a data warehouse is better suited for generating static operational reports and powering standard BI dashboards. The shape of the data also dictates the optimal platform. Data lakes accept almost any format, including raw videos, text notes, JSON files, and traditional tables. Data warehouses, however, require strictly curated, highly structured operational datasets.

These differences extend to their blueprint strategies. Data lakes utilizes a schema-on-read approach, meaning you organize the data only when you open it. Data warehouses rely on a schema-on-write model, where data must be cleaned and structured via ETL processes before it can enter the system.

Finally, financial considerations vary based on cost priorities. Data lakes offer highly cost-effective scaling for massive, diverse datasets. Conversely, data warehouses focus on maximizing ROI for highly structured, fast, and repeatable queries [Workday, 2026].

Question 4: What is a Data Mart? How does it relate to a data warehouse and who typically uses it?

Definition: According to Qlik (2026), a data mart is best understood as a highly structured, purpose-built data repository tailored specifically to the analytical needs of a particular department, line of business, or geographic region. Instead of forcing business units to dig through a cluttered, enterprise-wide database, these systems are typically carved out as

isolated, partitioned segments of a much larger corporate data warehouse. By focusing solely on a single subject area like isolated records for finance or sales teams, they allow everyday business users to run analytical queries and pull actionable insights significantly faster because they are working with a heavily scaled-down, deeply relevant dataset.

How Data Marts relate to a data warehouse?

When looking at a corporate data architecture, the relationship between these two tools comes down to scale and focus. Lu (2025) outlines that a data warehouse functions as a massive, centralized hub that integrates data from all corners of the company to establish a single, comprehensive source of truth. Conversely, a data mart is simply a specialized, smaller subset carved out of that warehouse to serve a targeted purpose [Lu, 2025].

Who Typically uses it?

Because data marts are lean and hyper-focused, Lu (2025) notes that they are ideal for department-focused teams like marketing operations or finance groups that need quick, targeted insights for their immediate daily tasks. On the flip side, full enterprise data warehouses are usually managed by core data engineering teams and utilized by high-level corporate analysts who require a holistic, historical look at the entire organization's performance [Lu, 2025].

Question 5: Explain the difference between Structured, Semi-Structured and Unstructured Data. Provide one example of each.

a) **Structured Data:** is defined as information that is highly organized and conforms to a set schema, allowing it to be easily queried using SQL (Snowflake Inc. 2026). **For example,** picture a perfectly neat and organized Excel spreadsheet, or a bank statement table. Everything is neatly recorded into predictable columns, such as Date of transaction, Description of a transaction and an Amount of the transaction.

b) **Semi-structured Data:** bridges the gap between unstructured documents and rigid databases by offering data modeling flexibility while maintaining an underlying organizational framework (Raveh 2024).

For example, think of an email, the body text is messy, but the "To" and "From" fields are organized.

c) **Unstructured Data:** is defined as messy, free-flowing information that does not follow a predefined format (or schema), because it lacks a rigid row-column structure, it is significantly harder to store, search, and analyze without specialized tools (Ademola-Shanu 2024).

For example, this is a raw and chaotic file such as videos, audio clips and a photo that have zero layout rules. This messy stuff actually makes up most of the data in existence today. A computer cannot quickly see what is inside the photo or hear what is said in the audio without using smart AI tools to decode it.

Question 6: What is Data Modeling? Why is it important when designing a system that works with data? Describe at least two types of data models.

Definition: Data modeling is a comprehensive process focused on building a visual representation of data and the specific relationships within it (Pyke, 2025). This practice works as a foundational blueprint that defines how data is structured, stored, and retrieved to maintain absolute consistency and clarity across data management environments (Pyke, 2025).

Why is it important in System Design: According to Olin (2025), Designing systems without data modeling often leads to miscommunication, operational inefficiencies, and ambiguity due to differing data interpretations. Data modeling provides a unified language and standard clarity that delivers critical enterprise benefits (Olin, 2025):

- System Optimization: Enforces the use of optimally formatted data to guarantee smooth execution of software, applications and business processes.
- Data Infrastructure: Directly guides and enables smooth data integration, structured storage, and efficient retrieval.
- Enterprise Alignment: Bridges communication gaps to improve collaboration between technical groups and business stakeholders.
- Platform Health: Maximizes system stability, long-term maintainability, and architectural adaptability while supporting data governance and quality compliance.

Types of Data Models:

As explained by Olin (2025), Data modeling is traditionally executed top-down across different levels of abstraction. Two primary types include:

1) Conceptual Data Model

- Core Purpose: Identifies and defines core business concepts alongside their overarching relationships to build a shared understanding among stakeholders.
- Characteristics: Operates at the highest level of abstraction, entirely independent of specific applications or underlying technologies, focusing purely on the big picture rather than granular detail (Olin, 2025).

2) Logical Data Model

- Core Purpose: Refines the conceptual model by adding precise details to satisfy specific business requirements without tying the design to a technical platform.
- Characteristics: Maps out entities, attributes, and explicit relationships while resolving many-to-many complexities using natural keys and bridge tables. Technical system details like index definitions and data types are explicitly excluded at this stage (Olin, 2025).

Question 7: What is Data Mining? How is it used in business? Provide two real-world examples of this in action? In the words of Chia (2024), data mining is the sophisticated extraction of hidden, potentially valuable information from vast, extensive datasets.

How is Data Mining Used in a business?

As outlined by Lee (2022), data mining functions as a powerful strategic framework that changes raw numbers into information vital for guiding core corporate operations. Businesses deploy data mining across various commercial workflows to drive distinct goals:

- Operational Efficiency: Pinpoints procedural flaws and forecasts hardware deterioration by linking internal server data with external metrics.
- Customer Journey Optimization: Tracks metrics like landing page duration and aftersales feedback to identify and resolve consumer friction points.
- Hiring Processes: Enhances human resource tasks by determining optimal candidate sources and formalizing step-by-step progress metrics.
- Inventory forecasting: Pinpoints seasonal marketplace variations and emerging buying patterns to protect companies against overstocking or understocking.

Two Simple Real-World Examples:

- a) The Diaper-Beer Connection (Market Basket Analysis)
 - The Method: Grocery chains look closely at customer receipts to spot patterns in buying habits that standard inventory tracking misses.
 - The Payoff: They noticed a strange trend where people buying diapers on Friday nights were highly likely to pick up beer at the same time. Shifting those two items closer together on the floor immediately drove up impulse sales.

b) Curated Feeds (Algorithmic Recommendations)

- The Method: Platforms like Spotify or Netflix monitor your daily activity not just what you play, but when you skip a track, pause a show, or loop a video.
- The Payoff: The system groups you with thousands of other users who interact with the app in the exact same way. It then looks at what they are obsessed with to feed you suggestions you didn't even know you wanted yet.

Question 8: What is a Database Management System (DBMS)? Name and briefly describe four popular DBMS platforms (e.g., MySQL, PostgreSQL, Oracle, MongoDB) and explain a suitable use case for each.

Definition: Database Management System.
As articulated by Knight (2024), refers to a coordinated integration of diverse software services designed to concurrently store, compute, maintain, organize and output data as a unified core product component.

Additionally, as explained by Hackscale (2023), a database management system is defined as software that is utilized to "define, manipulate, retrieve, store and manage data in database." It fundamentally acts as an interface layer in charge of defining data validation rules, updating records, analyzing information, and facilitating direct interactions between databases, applications and end users.

Popular DBMS Platforms and Their Use Cases:

DBMS Platform	Technical Description	Suitable Use Case
MySQL	A free, open-source relational database management system (RDBMS) owned by Oracle Corporation.	Database-driven web applications such as WordPress platforms, online forums, and content management systems.
PostgreSQL	A free, open-source RDBMS built originally as a successor to the Ingres database system.	Enterprise-grade applications requiring robust, open-source relational setups across varying OS platforms like Linux and Windows.
Oracle DBMS	A commercial, multi-model database system, recently rebranded as the Oracle AI database.	Heavy corporate operations running complex online transaction processing (OLTP) and highly-capacity data warehousing (DW).
MongoDB	An open-source, NoSQL document-oriented database designed to manage non-relational document structures.	Big data applications demanding intensive, high-volume data storage across distributed networks.

Question 9: What is ETL (Extract, Transform, Load)? Explain each step and describe why this process is important in data analytics.

Definition: As stipulated by Amazon Web Services (2026), Extract, Transform and Load (ETL) is defined as the foundational data integration process of combining raw information from multiple disparate sources and consolidating it into a centralized repository known as a data warehouse. The process utilizes a predetermined set of business rules to clean, validate, and systematically organize raw inputs, ensuring they are optimized for storage, predictive data analysis, and machine learning models.

Explain each step and describe why this process is important in data analytics:

As highlighted by PowerMetrics (2025), a standard data integration pipeline operates through three distinct stages:

a) **Extract:** The initial stage gathers raw payloads across distributed locations, including cloud databases, application programming interfaces (APIs), IoT devices, and unstructured systems like documents or social media feeds.

b) **Transform:** Extracted sets are parsed, checked for validation errors, and modified by applying business rules. This phase includes performing complex data calculation, structural enrichments, and running machine learning algorithms for error detection.

c) **Load:** The redefined data is permanently moved into target enterprise analytics systems (such as data lakes or data warehouses) utilizing either full database updates or incremental records loading.

Why is ETL important in data analytics?

According to Wang (2026), ETL is vital in data analytics because it converts raw, disorganized data into clean, consistent, and structured datasets that serve as the foundation for reliable business intelligence. Wang (2026) further explains that, a properly designed ETL pipeline eliminates operational errors, duplicates entries, and enriches data prior to warehouse storage, which directly "sets the stage for faster insights, better decisions, and smoother operations".

The author highlights 3 major organizational benefits that underscore the importance of ETL:

- Establishing a Unified View: ETL centralises disparate software and database sources to build a "consistent, complete foundation for analytics."
- Enhancing Data Quality: "Clean, standardize, and validate data" during transformation so only trusted data reaches the warehouse.
- Preserving Historical Context: ETL lets teams "retain and analyze snapshots of data over time" to track long-time trends.

Question 10: What are common data storage formats used in analytics (e.g., CSV, JSON, Parquet, Avro)? Briefly describe each and state when you would use one over another?

Data Storage format in analytics vary significantly by performance and compatibility, as detailed by Terry (2026). The primary formats include:

- CSV: is a good option for compatibility, spreadsheet processing, and human-readable data (Terry, 2026).

Temoy (2026) warns that it is inefficient and lacks nested data support.

- JSON: The author identifies JSON as being "heavily used in APIs" and ideal for small or landing datasets, noting that heavy nesting compromises its readability.

- Avro: As Temoy (2026) explains, this binary, row-based structure supports schema evolution, rendering it "ideal for row-level operations or in Karger-based Data lakes."

- Parquet: Characterized as a "columnar storage format," Temoy (2026) outlines its seamless integration with Hive and Spark for executing SQL queries.

- ORC: Temoy (2026) suggests that while less popular than Parquet, ORC offers "better compression and better schema evolution support" in big data frameworks.

Comparing the 4 Main Formats:

Format	Mechanism	Trait	Best For
CSV	Text (Rows)	Comma-separated tables	Fast Excel sharing
JSON	Text (Key-Value)	Labeled nested structures.	Web APIs and config files.
Parquet	Binary (Columns)	Column-block organization.	High-speed warehousing queries.
Avro	Binary (Rows)	Self-contained Schema records	Live streaming and event logging

When to Choose Which?

- Opt for Parquet when crunching massive enterprise metrics, like calculating row-row averages or total across millions of data points, because it isolates and pulls only the exact columns required for a query; processing speeds are incredibly fast.
- Stick with CSV or JSON if the file needs to be directly opened, reviewed, or tweaked by a person without spinning up complex code or developer tools.
- Lean toward Avro for live, sub-second data streaming feeds. It allows the systems to append fresh records sequentially, line by line, keeping ingests fast without bottlenecking operations.

Question: ||

What is Artificial Intelligence (AI)? Define it and explain the difference between Narrow AI (Weak AI) and General AI (Strong AI). Provide one example of each.

The data education center documentation (2026) establishes a clear distinction between the task-specific frameworks applied in the modern digital landscape and the conceptual models governing future computational intelligence.

Definition: Artificial Intelligence is broadly characterized as a transformative technological mechanism designed to restructure "how we live, work, and interact with technology". At its core, the technology involves developing digital architectures capable of "learning and adaptation", data processing, and replicating specific components of human cognition to execute complex diagnostic or computational problems. (The data education center docu-

Narrow AI Vs General AI

Criterion	Narrow AI (Weak AI)	General AI (Strong AI)
Scope	Single-task focus.	Universal human capabilities.
Status	Active globally today.	Completely conceptual.
Analogy	A standard calculator.	A flexible human brain.

Simple Examples:

- a) Narrow AI: FaceID or Spotify recommendations.
- Action: Unlocks your device by scanning your face instantly.
 - Limit: Specialized in facial mapping but cannot write essays, play chess, or navigate vehicles.
- b) General AI: JARVIS (Iron Man) or sci-fi androids.
- Action: Completely self-aware, multifunctional digital minds.
 - Limit: Hypothetical; possesses the cognitive flexibility to cook, write books and learn instruments simultaneously.

Question 12) What is Machine Learning (ML)? How does it differ from traditional programming? Describe the 3 main types of Machine Learning: Supervised, Unsupervised and Reinforcement Learning.

Definition: Chen (2024) explains Machine Learning (ML) as a specialized subset of artificial intelligence (AI) that focuses on building systems that learn and improve automatically as they consume more data.

How does it differ from traditional programming?

The OTC Team (2026) notes that the fundamental distinction between machine learning (ML) and traditional programming lies in how operational rules are defined and how data is processed. In traditional software engineering, a human developer manually writes explicit, logic-based instructions for the computer to follow. In contrast, machine learning allows systems to independently discover patterns and generate their own rules by analyzing historical datasets.

Machine Learning Vs Traditional Programming

Feature

Logic Creation

Traditional Programming

Explicitly coded by a human software engineer

Machine Learning

Autonomously learned from input data.

System Workflow

Input + Rules + Output
(Executes fixed formulas)

Input + Output →
Algorithm (Creates the model).

Adaptability

Completely static; requires manual code updates.

Inherently dynamic; improves with more data exposure.

Task Suitability

Ideal for predictable, rule-based tasks
(e.g., payroll)

Ideal for uncertain, data-heavy tasks
(e.g., fraud detection)

Three main types of Machine Learning:

Supervised, Unsupervised, and Reinforcement Learning:

Syracuse University School (2025) highlights three dominant instructional paradigms utilized within the domain:

a) Supervised Learning: The model functions under the guidance of a predefined dataset where correct answers are labelled, mimicking a student learning from a tutor.

b) Unsupervised Learning: The system evaluates completely unlabeled datasets to reveal hidden relationships and sort variables organically by matching relative structural similarities or differences.

c) Reinforcement Learning: An interactive trial-and-error approach where an algorithmic agent explores its environment, processing environmental rewards or penalties to organically maximize positive strategic outcome over time.

Question 13: What is Deep Learning? How does it relate to Machine Learning and Artificial Intelligence? What are Neural Networks and how do they work?

The International Organization for Standardization (ISO 2026) explains Deep Learning (DL) as a specialized subset of machine Learning and artificial intelligence designed to train computers to replicate the way biological neurons in the human brain process information. The author notes that by utilizing artificial neural networks (ANNs) that simulate multiple computational layers, machines can uncover intricate insights hidden within massive data volumes and independently complete tasks that

once required human talent.

How does Deep Learning relate to Machine Learning and Artificial Intelligence?

Amazon Web Services (2026) explains that Deep Learning (DL), Machine Learning (ML), and Artificial Intelligence (AI) are structurally nested within one another, with deep learning serving as a highly specialized subset of machine learning, which in turn acts as a subset of artificial intelligence.

The author highlights that these methodologies collectively exist to complete complex computational tasks that would otherwise demand vast amounts of time and development resources if executed through traditional manual programming methods.

The Nested Structural Hierarchy:

The author details the explicit relationships and individual boundaries of the three domains as follows:

- Artificial Intelligence (AI): Represents the broadest data science envelope framing both sub-disciplines.
- Machine Learning (ML): Functions as the core science of training computer systems to process large datasets, organically identify patterns, and predict outcomes without explicit, hardcoded rules.
- Deep Learning (DL): Operates as an advanced, deep-layered expansion of ML modeled directly after the human brain's neural structures. It is specifically deployed to automate highly abstract tasks such as natural language processing or image analysis that traditionally depend on human intelligence.

What are Neural Networks and How Do They Work?

Lee (2026) explains that a neural network is a predictive machine learning model that stacks simple artificial "neurons" in layers to learn pattern-recognizing weights and biases from data, effectively mapping input vectors to distinct outputs. The author notes that what separates neural networks from traditional machine learning algorithms is this unique, layered architecture and its innate capability to execute complex, nonlinear transformations.

Lee (2026) further explains that a standard neural network uses 3 main layers to process data:

- Input Layer: Receives the raw features from the dataset.
- Hidden Layers: Use math to multiply the incoming features by specific weights (importance levels) and add a bias (trigger threshold).
- Output Layer: Applies a calculation to turn those results into a final prediction or percentage.

Question: 14 What is Natural Language Processing (NLP)? Describe what it is and provide 3 real-world applications of NLP that you use in everyday life. (Robtree (2024) defines Natural Language Processing (NLP) as a specialised branch of artificial intelligence (AI) that centres on the direct interaction between computers and humans using natural language. The author notes that the primary objective of this field is to program computer systems to efficiently process and evaluate massive quantities of

natural language datasets.

Three real-world applications of NLP that we use in everyday life:

a) Predictive Keyboard Mechanisms:

- Functional Logic: Mobile devices monitor current sentence context to anticipate upcoming vocabulary while seamlessly rectifying spelling errors via statistical language models.

b) Semantic Search Engineering:

- Functional Logic: Modern web queries look beyond literal vocabulary to parse user intent. For instance, inputting "indoor dining spots during rain" prompts the system to isolate local covered restaurants rather than pulling up generic weather articles.

c) Vocal Command Automation:

- Functional Logic: Conversational assistants decode audible speech into machine-readable text, isolate the target request, retrieve relevant information, and synthesize a fluid vocal output.

Question 15: What is a Large Language Model (LLM)? How is it different from a standard Machine Learning model? Name two well-known LLMs and explain what they are used for.

Luna (2023) defines a Large Language Model as an advanced artificial intelligence system specifically engineered to model and process human language. The author stipulates that these models earn the

designation of being "large" due to their massive structural complexity, typically consisting of hundreds of millions or even billions of parameters that dictate the system's operational behaviour and patterns.

How Large Language Model Differ From a Standard Machine Learning Model?

According to The Toloka Team (2026), the core difference between Large Language Models (LLMs) and traditional Machine Learning (ML) models is determined by their structural scale, data requirements, and operational flexibility. While standard machine learning represents a broad domain where systems automatically extract patterns from data, an LLM is a highly specialized application nested deep within ML that focuses exclusively on understanding and generating human language via a deep learning framework.

Traditional ML Vs LLM

Feature	Traditional Machine Learning	Large Language Model
Scale	Thousands to millions of parameters.	Billions to trillions of parameters.
Data	Small, task-specific datasets.	Massive, internet-scale text.
Flexibility	Single, dedicated function per model.	Many diverse tasks via prompting.
Cost	Hours to days; minimal budget.	Weeks to months; millions of dollars.
Output	Discriminative (labels/classifies inputs).	Generative (Creates new text).

Two Well-known LLMs and Their Uses

The Databricks staff (2026) identifies several prominent options within the crowded field of artificial intelligence, explicitly listing Claude and the Generative Pretrained Transformer (GPT) family as two well-known Large Language Models (LLMs).

The author outlines that these advanced models are trained on massive datasets to learn the underlying patterns of human language, allowing them to surpass previous technologies in performing highly complex, context-aware tasks.

As articulated by Belcic and Stryker (2026), Claude and the Generative Pretrained Transformer (GPT) family represent prominent generative artificial intelligence models built on a transformer deep learning layout. The authors establish that these foundational large language models (LLMs) dynamically process raw data patterns to mimic human-created outputs across diverse application tracks.

System Profiles:

The Unique Backgrounds of these 2 Systems:

a) Generative Pretrained Transformer (GPT): Built by OpenAI, this series scaled from GPT-1 in 2018 to GPT-4, which uses around 1.8 trillion parameters. Belcic and Stryker (2026) point out that its multimodal version, GPT-4o, handles text, audio, and video together instantly.

b) Claude: The text identifies this separate platform as a major competing model created by Anthropic.

Question 16: What is Generative AI? How does it work at a high level? Name 3 tools that use

Definition:

Generative AI and describe what each one does. The Cape Peninsula University of Technology (CPUT 2026) explains Generative AI (GenAI) as a specialized branch of artificial intelligence design to synthesize entirely new content such as text, images, music, and videos by processing diverse input modalities, including computer code, voice and images.

How it works at a high level?

The Singapore Management University Academy (SMU Academy 2026) elaborates that generative AI functions by systematically learning structural patterns from large datasets and leveraging those rules to synthesize entirely new digital outputs.

3 stages workflow:

3 Tools that use GenAI and what each one does:

Stryker and Scapicchio (2026) identify ChatGPT, DALL-E, and Midjourney as leading generative tools that transform user prompts into original content combinations through deep learning architectures:

- ChatGPT: Operates as a text-generation application that handles dialogue, answer open-ended inquiries and composed written assets like essays or papers.

- DALL-E: Functions as diffusion-based image creation tool that removes visually noise to build realistic illustrations from text descriptions.

- Midjourney: Uses transformer networks and attention mechanisms to interpret user instructions and generate highly complex digital art and designs.

Question 17: What is a Training Dataset? Why is the quality of training data so important when building an AI model? What could go wrong if the training data is biased or incorrect?

Definition: H2O.ai (2026) defines a training dataset (or training set) as the baseline example data fed into machine learning (ML) algorithms during a model's building phase. The publication highlights that these datasets serve to teach the algorithms how to autonomously formulate predictions and extract meaningful hidden insights from within the supplied data matrix.

Why the Quality of Training Data Matter?

Mercor (2026) states that training data dictates how effectively an AI model generalises, manages edge cases, and produces reliable outputs. Under the principle of "garbage in, garbage out", flaws in the training data permanently corrupt model behaviour.

What Can Go Wrong If Data Is Biased or Incorrect?

Rogers and Jonker (2024) warn that utilizing biased or unrepresentative training data can severely disrupt the behaviour of AI systems, resulting in unfair, inaccurate, and unreliable models.

Key Risks of Data Bias:

Rogers and Jonker (2024) demonstrate that skewed data inputs corrupt model behaviour across 4 primary domains:

- a) Discrimination: Skewed data forces algorithms to mirror historical inequalities. For example, automated recruitment systems over-indexing male profiles directly disadvantage qualified female applicants.
- b) Warped Decisions: Algorithms trained on unrepresentative parameters deliver faulty predictions, causing businesses to misread market dynamics and misallocate resources.
- c) Legal Penalties: Deploying biased algorithms under rules like the EU AI Act can incur corporate fines scaling up to €5 million or 7% of global revenue.
- d) Feedback Loop: Re-ingesting biased output patterns as new training inputs permanently locks in the original systemic distortions over time.

Question 18: What is Computer Vision? How do machines 'see' and interpret images? Provide 2 real-world use cases of computer vision.

Byjites: As formulated by (Abellar and Stryker (2026), computer vision is a specialized subfield of AI that equips machines with the functional ability to process, analyze and interpret visual inputs like images and videos.

How machine "see" images? :

Snowflake (2026) outlines that computer vision platforms process raw pixels through 4 distinct steps:

1) Ingestion and Preprocessing: Captures raw images from sensors or cameras, then normalizes scale, filters

noise, and adjusts for lighting changes.

2) Feature Extraction: Employs mathematical operations to isolate core visual markers like edges, shape, colours and textures.

3) CNN Classification: Uses convolutional neural networks (CNNs) to match these features against learned definitions, building from basic lines to complete objects for real-time identification.

4) Feedback Refinement: Implements active feedback loops that flag misidentified targets and re-route them as new training data to steadily boost accuracy.

Two Real-World Examples:

Snowflake (2026) notes that computer vision transforms raw pixels into real-world action:

a) Autonomous Vehicles: Self-driving cars deploy deep learning to detect pedestrians, street signs, and lane lines instantly. Cities use this data to adjust traffic grids and optimize road safety.

b) Healthcare Diagnostics: Image models flag anomalies, locate tumors in X-rays.

Question 19: What is an API (Application Programming Interface)? Explain what it is using a simple real world analogy. Why are APIs important in modern software and data systems?

Definition: Amazon Web Services (2026) explain that an API acts as an operational mechanism designed to enable 2 separate software components to communicate with each other using an established set of definitions and protocols.

Simple real-world analogy:

The Restaurant Analogy
Goodwin (2026), explain the interaction between systems using a simple service model:

- a) The client (The customer): You sit at a table looking at a menu, representing an application requesting specific information of a service.
 - b) The server (The Kitchen): The kitchen holds all raw ingredients and tools to prepare your order, but you cannot access or view it directly.
 - c) The API (The Waiter): The waiter carries your order to the kitchen, tells the chefs what to do, and returns to deliver your meal (the data response).
- Why APIs are vital today?

Kulchshkina (2025), establishes that APIs function as the core framework of modern software engineering, enabling multi-platform connectivity.

workflow automation, and real-time integration across cloud systems.

Kulkushkina (2025) outlines that APIs serve as the core framework of modern software engineering, driving operational efficiency through capabilities:

a) Interoperability: They enable distinct applications, databases, and IoT components to communicate seamlessly in real time.

b) Data control: They allow platforms to fetch, update, modify and erase records across different systems instantly.

c) Automation: They link independent tools together to run automated workflows, eliminating manual data-entry errors.

d) Integration: They let engineers quickly plug in third-party services like payment gateways without coding from scratch.

e) Microservices: They act as separate structural paths, letting teams update specific app features without crashing the infrastructure.

Question 20: What is an API Key? What is its purpose? How should it be stored and protected? and What could happen if it's exposed publicly?

Definition: Jackson and Goodwin (2025) define an API as a unique alphanumeric string used to identify and verify a calling application before a server grants access via an application programming interface (API).

What is its purpose?

Core Purpose and Intent:

Jackson and Goodwin (2026) state that the primary purpose of an API key is to monitor API calls, manage consumption, and ensure adequate bandwidth. They enable organizations to block unauthorized traffic and enforce rate limiting parameters to control requests frequency.

How to Store and Protect it?

Because keys are often generated ~~in~~ⁱⁿ plain text, their security depends on proper storage:

- a) Secure Vault Storage: Professionals recommend storing keys as hashed values in a database or using dedicated secrets management systems for production apps.
- b) Decoupling from Source Code: Avoid embedding keys directly in source code to prevent public exposure during publication.
- c) Key Rotation: Routinely rotating and generating new keys every 90 to 180 days helps maintain security.

Consequences of Public Exposure:

Jackson and Goodwin (2026) emphasize that anyone with access to an API key can use it, much like a password. A leaked key can lead to unauthorized usage, data exposure, service abuse, and reputational damage.

Question 21: What is a REST API? What does REST stand for and what are the main HTTP methods (Get, Post, Put, Delete) used in REST APIs? Briefly describe each method.

Definition: Amazon Web Services (2026) defines REST API (or REST API) as an operational interface that distinct computer systems utilize to securely and efficiently exchange information over the internet. The author notes that these interfaces are built to support data-driven transactions between internal and third-party corporate applications because they adhere to highly structured, reliable software communication standards.

What does it stand for?:

According to Amazon Web Services (2026), REST stands for Representational State Transfer.

In discussing this architectural framework, the author emphasizes that it acts as "a software architecture, that imposes conditions on how an API should work" (AWS, 2026).

What are the main HTTP methods (Get, Post, Put, Delete) used in REST APIs? Briefly describe each method:

Think of the commands/methods like managing files on your computer:

- a) **GET (View):** Grabs data from the server without changing anything, like browsing a friend's profile.
- b) **POST (Create):** Sends new data to the server to build something fresh, like publishing

a new post.

c) PUT (Replace): Completely overwrites existing data with new info, like updating your entire account bio.

d) DELETE (Remove): Wipes data from the server permanently, like clicking "Delete Photo".

Question 22: What is JSON (JavaScript Object Notation)? Why is it commonly used when working with APIs and data exchange? Write a simple example of a JSON object.

Definition: Based on Amazon Web Service (2006), a JavaScript Object Notation (JSON) is defined as an open data interchange format structured to be both human and machine-readable.

Why is JSON commonly used for APIs and data exchange?

According to the ITU Online Editorial Team (2024), the following ^{are} key reasons for JSON widespread in the industry ~~and~~ or its adoption:

a) Compactness and Speed: The authors point out that JSON reduces payload sizes and overhead compared to XML.

b) Language Independence: The text underscores that JSON works universally across platforms like Python, Java and Go.

c) Dual Readability: The team highlights that it is "easy for people to read and easy for machines to parse".

Simple Example:

Based on IBM (2024), a JSON object is structured as a collection of comma-separated name/value pairs enclosed in curly braces. The author emphasizes that this format cleanly groups distinct data types for rapid network processing.

```
{  
  "givenName": "John",  
  "surname": "Doe",  
  "employeeId": 10452,  
  "isActive": true  
}
```

Question 23: What is Prompt Engineering? Why is it an important skill when working with AI tools like ChatGPT or Claude? Provide 2 examples of a poorly written prompt and an improved version of each.

Definition: According to the Databricks Staff (2024), prompt engineering is an emerging AI field focused on designing and optimizing text inputs to guide generative models toward user goals.

Why is Prompt Engineering an important skill when working with AI tools like ChatGPT or Claude?

Based on Mitchell (2025), prompt engineering is the practice of designing inputs to guide AI models.

The author emphasizes that the process "involves crafting, testing, and refining instructions" to optimize performance across platforms like ChatGPT and Claude.

The text highlights 4 main reasons why this skill is important:

a) Tailoring Behaviour: Distinct models process language differently, requiring structured modifications.

b) Efficiency: Clear instructions up front save processing time and cut down on tedious trial and error.

c) Accessibility: It allows non-technical users to easily navigate complex AI tools with consistent results.

d) Risk mitigation: Responsible design safeguards against systemic errors, bias, and misinformation.

Provide 2 examples of a poorly written prompt and an improved version of each.

Graderha (2026) outlined that, a poorly written, unconstrained prompts lack the strict context or parameters required to guide large language model effectively.

Example 1 (Few-Shot Strategy):

- Poor: "Explain climate change"

- Improved: Provide targeted formatting examples first, then insert: "Now explain: Climate Change"

- Poor: "Write a summary about climate change, its causes, and its environmental effects."
- Improved: Break the query down explicitly:
"Step 1: Define... Step 2: Explain... Step 3: Describe..."

Question 24: What is a MAP function? Explain what it does in programming and data processing. Provide a simple example in either Python or JavaScript.

Definition: According to Kumar ~~and~~ writing for DataCamp (2022), Python's `map()` function is a powerful built-in tool that enables functional programming patterns for data transformation, making it highly effective for cleaning text data, processing large datasets, or executing bulk calculations.

What it does?

As explicitly stated by Sr (2022), the syntax `map(function, iterable)` yields a flexible "map object" easily converted back into a standard list. The author underscores that the function scales seamlessly across data tasks:

- Single data sets: Sr (2022) demonstrates mapping functions over a lone list to calculate squared values instantly.
- Multiple data sets: It is argued by the author that the tool processes parallel streams by accepting multi-argument functions.
- Inline logic: The author points out that developers

can run these operations on the fly using anonymous lambda functions.

Simple Python Example:

Simple variable assignment and math execution:

python

```
# simple variable assignment and math execution
first_number = 10
second_number = 5
total_sum = first_number + second_number

print ("The sum is:", total_sum)
```

Question 25: What is cloud computing? Name 3 main cloud service models (IaaS, PaaS, SaaS) and briefly explain each. Give one example of a product for each model.

Definition: Sunjara and Malley (2026) explains cloud computing as the on-demand access to virtual or physical computing resources over the internet, utilizing a flexible pay-per-use pricing model.

3 main cloud service models (IaaS, PaaS, SaaS)

Amazon Web Service (2026) outlined 3 main cloud service models and their real-world product examples as follows:

a) IaaS: Amazon Web Service (2026) points out that IaaS offers basic virtual compute, networking, and storage building blocks, granting users the highest level of resource control.
Example: Amazon EC2.

c) PaaS: It is argued by the author that PaaS removes the need to manage underlying operating systems, allowing developers to focus purely on application deployment. Example: AWS Elastic Beanstalk.

d) SaaS: The author notes that SaaS delivers fully operational end-user applications managed entirely by the vendor. Example: Web-based email.

Question 26: What is Version Control System? What is Git and why is it widely used by data analysts and developers? Briefly explain what a 'commit' and a 'branch' are.

Definition: As explicitly stated by Atlassian (2025), a version control system acts as a persistent database that meticulously tracks every single modification made to the source code codebase.

What is Git and why it is widely used by data analysts and developers?

Definition: According to Horsley (2023), Git is the world's most popular version control framework and a critical asset for software engineering and data analytics teams.

Worsley (2023) points out that it functions as a critical daily tool due to several key performance factors, and it is widely used by data analysts and developers:

- a) High operational speed: The author highlights that keeping full local copies makes workflows exceptionally fast, bypassing constant server communication delays.
 - b) Offline workflow capabilities: It is argued that by the author that team members can freely review code and edit files offline, requiring internet access only when pushing final commits.
 - c) Built-in fault tolerance: The author notes that tracking every file alteration creates a reliable safety net, allowing teams to instantly revert to older versions if errors occur.
 - d) Parallel branch collaboration: Author points out that contributors can build separate features in isolated branches and merge them cleanly, avoiding developer code conflicts.
- What is a 'Commit' and a 'Branch'?

As explicitly stated by the ITU Online Editorial Team (2026), these terms serve distinct operational roles:

- a) Commit: The author underscores that a commit permanently logs a snapshot of code modifications at a single point history.
- b) Branch: Allows teams to maintain an independent line of development without creating a

duplicate copy of the project folder.

Question 27: What is a Jupyter Notebook?

What programming language is it most commonly used with? Why is it popular among Data Analysts and Data Scientists?

Definition: A Jupyter Notebook is a browser-based, interactive environment that integrates execution, code, data visualisations, markdown text, and ~~html~~ inline media in a single document. As explained by the Nebius Team (2025) this framework operates technically as a JSON file using a .ipynb format that communicates with a background kernel to manage runtime memory states.

What programming language is it most commonly used with?

Ashwani (2024) highlights the following details:

a) The dominant ecosystem: Python is the most common programming language used within the ecosystem, executed via the standard built-in Jupyter kernel layer.

b) Multi-language Adaptability: It is argued by the author that, despite the heavy prevalence of Python, the tool maintains native versatility by supporting alternative data science kernels like R and Julia.

c) Persistent session states: Whichever language is selected, the underlying kernel maintains variables and data structures across independent cells so users don't have to re-run previous scripts.

Why Data Teams Rely on it? :

- a) Granular, Interactive Execution: Instead of executing a sprawling script, analysts run isolated code chunks sequentially, validating data pipelines instantly.
- b) Inline Visualization Framework: Plots and data tables render directly beneath the active cell, letting you spot trends without window-switching.
- c) Cohesive Technical Narratives: Operating like a research log, it embeds program logic into natural text to simplify stakeholder presentations.

Question 28: What is Python? Why is it considered one of the most popular programming languages for Data Analytics and AI? Name 4 Python libraries commonly used in data work and briefly describe each.

Definition: Python is a highly high-level, object-oriented programming language characterized by an easy-to-interpret runtime model and highly readable syntax (Oracle, 2024).

Why is it considered one of the most popular programming languages for Data Analytics and AI?

As stated by Kleemann (2023), Python reliably handles every stage of a data pipeline—from initial collection to production scaling.

- a) Background detail abstraction: The language removes tedious low-level machine tasks, allowing teams to get straight to data analysis.
- b) Robust infrastructure capabilities: The ecosystem

handles diverse database structures and environment isolation tools like pip and conda seamlessly.

c) Advanced visual storytelling: Dynamically developing visualization libraries like Plotly and Seaborn simplify communicating complex patterns to business stakeholders.

d) Scalable cluster integration: The language integrates naturally with frameworks like PySpark and Dask to facilitate distributed big data analytics.

Python libraries commonly used in data work:

Ali (2025) highlights 4 core libraries that form a backbone of modern data operations:

a) Pandas: A powerful tool used for cleaning data and organizing datasets into structured DataFrame tabs.

b) Numpy: A high-speed computing library engineered to perform complex mathematical and array calculation efficiently.

c) Matplotlib: A comprehensive charting framework built to turn raw datasets into diverse visual plots and graphs.

d) Scikit-Learn: An industry-standard platform that provides user-friendly tools to train and deploy machine learning models.

Question 29: What is Data Privacy? Why is it important? Research the GDPR and explain what it is and which countries it applies to.

Definition: As stated by Kosinski and Foray (2026), Data Privacy, also called "information privacy" is the fundamental principle that indivi-

Individuals must retain direct authority over their personal data, including the right to dictate how organizations collect, store and process this information.

Why is it important?

a) Safeguarding Personal Integrity: It protects financial records, medical histories, and communications from unauthorized access.

b) Preventing Fraud: Proactive privacy mitigates identity theft vulnerabilities to prevent financial loss and emotional distress.

c) Preserving Trust and Corporate Reputation: Ethical data practices build strong customer trust and protect against data leaks.

d) Regulatory Compliance: Robust frameworks ensure adherence to laws like GDPR and CCPA, avoiding severe financial penalties.

What is the GDPR?

According to gdpr.eu (2026), The General Data Protection Regulation (GDPR) is a comprehensive data privacy and security law enacted by the European Union (EU) to regulate how personal data is collected, stored, and processed. It was put into effect on 25 May 2018; it established strict rules regarding user consent, data security and individual privacy rights, levying severe financial penalties against organizations that violate its standards.

Countries it apply to?

It applies to all EU member states as well as global organizations collecting data from individuals.

residing in the EU.

Question 30: What is Data Ethics? Provide 3 examples of unethical use of data or AI and explain what makes each example problematic.

Definition: Data ethics is a specialized field of ethics that governs the principles behind how organizations gather, protect, process and utilize information. It focuses on the fundamental moral obligations that entities have when collecting and disseminating personal information about individuals (Cepetai, 2024).

3 Examples of Unethical Data or AI use:

As outlined by SAS Institute Inc. (2026), 3 prominent examples of unethical use of data or AI highlight severe socio-technical flaws:

a) Algorithmic Bias in Healthcare: Systems use skewed data. This caused unequal care for marginalized groups.

b) Opaque Underwriting in Insurance: Algorithms settle claims without explanation. This destroys consumer trust.

c) Invasive Data Ingestion in Banking: Models ingest sensitive personal data without consent. This violates user privacy.

Question 31: What is Data Governance? Define it and explain the key components of a good data governance framework.

Definition: According to Databricks (2026), Data governance is a holistic framework encom-

passing the core principles, practices and technologies an organization deploys to manage its operational data assets across their entire lifecycle.

Key Components of a Good Data Governance Framework

As explained by Amazon Web Services (2026), good framework requires the following components:

- a) Data Curation: Automated processes limit redundant copies and identify sensitive data. This keeps information accurate for business use.
- b) Centralised Cataloguing: Metadata is organized into a single repository. This allows users to find and understand data assets easily.
- c) Access Control: Granular security protocols are enforced across the business. This balances data privacy with user accessibility.
- d) Continuous Monitoring: User and AI interactions with data are tracked. This ensures compliance and simplifies audits.

Question 32: What is Bias in AI? Explain what it means for an AI model to be biased. Provide a real world example of AI bias and describe the potential consequences.

Definition: As established by SAP (2024) Bias in AI refers to the occurrence of systematic errors or prejudices within AI systems that lead to distorted, inaccurate, or unfair outcomes for specific groups of people.

What it means for a Model to be Biased?

An AI model is biased when its mathematical architecture consistently produces distorted, unfair or inaccurate outputs against specific groups of people. This happens when the algorithm relies on faulty assumptions or structurally flawed data parameters (Holdsworth, 2023).

a) Prejudice and Stereotyping Bias: The algorithm converts historical societal inequalities or systemic stereotypes into rigid predictive rules.

b) Measurement and Sample Bias: The model evaluates incomplete or unrepresentative training data that fails to capture the real-world population.

c) Confirmation and Exclusion Bias: The system over-indexes on historical trends or fails to incorporate vital baseline factors.

Real-world Example and Potential Consequences

a) Recruiting Disparities: When an enterprise trains a hiring algorithm on historically male-dominated resumes, the system isolates gender as a proxy metric for success. It then actively penalizes female applicants by filtering out applications that mention women's organizations.

b) The Consequences:

— Socioeconomic Exclusion: Historically marginalised demographics face systemic barriers that hinder their ability to participate in the economy.

— Degraded Corporate Performance: Diverse teams lose the capacity to make sound analytical choices, which reduces overall accuracy and eliminates

Erosion of Public Trust: Discriminatory automated outcomes generates deep public backlash and foster permanent consumer mistrust.

Question 33: What are the key differences between a Data Analyst, a Data Scientist, a Data Engineer, and an AI Engineer? Create a simple table comparing their roles, skills and tools.

a) Data Analyst: Act as business detectives. They analyze past data to see what went right or wrong. This turns spreadsheets into insights for everyday decisions.

b) Data Scientist: They serve as the forecasters. They look at messy data patterns. They use math to predict future customer behaviour and trends.

c) Data Engineer: They function as digital architects. They do not analyze data. They build and maintain pipelines to store raw information securely.

d) AI Engineer: They operate as the implementers. They use smart machine learning models. They stitch these models into software to automate capabilities.

Quick Simple Comparison Table

Role	Objective / Goal	Skills	Tools
Data Analyst	Business Intelligence Metric	Finding trends	Excel, Power BI
Data Scientist	Statistical forecasting	Advance Math	Python, Statistics
Data Engineer	Pipeline architecture	Building code pipe	Cloud Systems
AI Engineer	Production model deployment	Software building	OpenAI, Python

Question 34: What is BI? Name 2 popular BI Tools (eg, Power BI, Tableau) and explain how they help businesses make data-driven decisions.

Definition: Edmond and Crabtree (2026) highlight the Business Intelligence (BI) is an enterprise practice focused on collecting, structuring, and analyzing data to uncover actionable insights that help organizations outpace competition.

2 Popular BI Tools:

a) Microsoft Power BI: Power BI provides highly scalable reporting built for Microsoft Fabric. It creates visual reports automatically from plain English text prompts.

b) Tableau: Delivers great visual design flexibility for deep data exploration. It handles huge datasets easily and flags unexpected changes automatically without manual dashboard builds.

How they help the business

a) Microsoft Power BI:

- Natural Language Queries: Allow managers to generate visual data reports instantly via plain English text prompts.

- Native Architecture Integration: Reduces corporate infrastructure costs by running analytics natively inside a company's existing Microsoft Fabric setup.

b) Tableau:

- Automated Risk Mitigation: Prevents business

blind spots by using automated AI scanning to immediately flag metric anomalies.

- High-Performance Exploration: Improves strategic analysis by allowing users to explore massive datasets deeply without experiencing system slowdowns.

Question 35: Which concept from this assignment did you find most interesting or surprising? Explain why?

* It was AI Bias. It was interesting because it is shocking to learn that AI is not as neutral as we think, it often repeats the real-life hidden prejudices in information humans give it.

Question 36: Which concept did you find most difficult to understand? What resources did you use to help you understand it better?

* It was Version Control systems and Git. What made it tough was that Git requires a completely different mindset. Wrapping my head around those abstract concepts felt incredibly confusing at first. I used sources from the internet for a better understanding.

Question 37: How do you see Data and AI affecting your future career or everyday life?

* Data and AI are most definitely going to reshape my career and daily routines by enhancing my technical skills and do tasks differently. I will be able to ditch routine work by using my skills and making smart choices instead of a repetitive work. This will also save me time.